

Nonextractable polyphenols, usually ignored, are the major part of dietary polyphenols: a study on the Spanish diet.

SCOPE: Dietary polyphenols (PP) can be divided into two groups: extractable polyphenols (EPP) or compounds solubilized by aqueous organic solvents, and nonextractable polyphenols (NEPP) or compounds that remain in their corresponding extraction residues. Most studies on food polyphenols and dietary intakes address exclusively EPP. The objective of this work was to determine the actual amount of PP, including NEPP, in food and in a whole diet. METHODS AND RESULTS: HPLC-MS analyses were performed to identify EPP in methanol-acetone extracts and NEPP in the acidic hydrolyzates of their extraction residues in cereals, fruits, vegetables, nuts, and legumes. NEPP contents, estimated as hydrolyzable PP plus nonextractable proanthocyanidins (PA), ranged from 880 mg/100 g dry weight in fruits to 210 mg/100 g in cereals and were substantially higher than the contents of EPP. NEPP intake (day/person) in the Spanish diet (942 mg) is higher than EPP intake (258 mg). fruits and vegetables (746 mg) are the major contributors to the total PP intake (1201 mg). CONCLUSION: Non extractable polyphenols are the major part of dietary polyphenols. The knowledge of intakes and physiological properties of NEPP may be useful for a better understanding of the potential health effects of dietary PP.